Facultatea de Electronică, Telecomunicații și Tehnologia Informației

Technical University of Cluj-Napoca

ETTI

Soil Moisture-Based Irrigation System

Using Resistive Moisture Sensors and Hysteresis Comparator

Student

Nițescu Gabriel-Nicolae

Coordinator

Raul Traian Fizeșan

Facultatea de Electronică, Telecomunicații și Tehnologia Informației

Contents

1.Goal and Significance

The aim of this project is to create and execute a mechanized irrigation system that sustains ideal soil moisture levels for plants. This is accomplished by utilizing a soil moisture sensor to detect the levels of moisture and an electronic control system to activate a water pump based on the readings from the sensor.

Automated irrigation systems aid in water conservation and provide optimal water distribution to plants, hence promoting growth and minimizing the need for manual intervention. The objective of this project is to showcase the practicality of implementing such a system by utilizing easily accessible electronic components.

Given that the resistance of the sensor varies between 750kΩ and 100kΩ as soil humidity changes from 55% to 85%, this formula can be used to determine the decrease in resistance corresponding to a 1% increase in humidity.

$$\Delta\mathrm{R1\%} = \frac{750k\Omega - 100k\Omega}{90 - 20} = 9.28k\Omega$$

The values of the resistance at 55% and 85% humidity can be determined using these formulas.

$$R55\% = 750 - 9{,}28(55 - 20) = 425.2k\Omega$$

$$R85\% = 750 - 9{,}28(85 - 20) = 146.8k\Omega$$

To continue, the corresponding voltage level for the 750kΩ resistance of the sensor is 9V, and for the 100kΩ resistance, it is 2V. The threshold voltage values were determined using the following formulas.

$$\mathrm{VThH} = \frac{10\% * R55\%}{750k\Omega} = \frac{4252}{750} = 5.67V$$

$$\mathrm{VThL} = \frac{10\% * R85\%}{100k\Omega} = \frac{1468}{100} = 1.95V$$

2.Block Diagram

To better understand, the designed circuit will first be displayed as a block diagram, which will be covered in the following pages. Finally, the complete circuit will be presented. The structure of the circuit is shown in Figure 1.

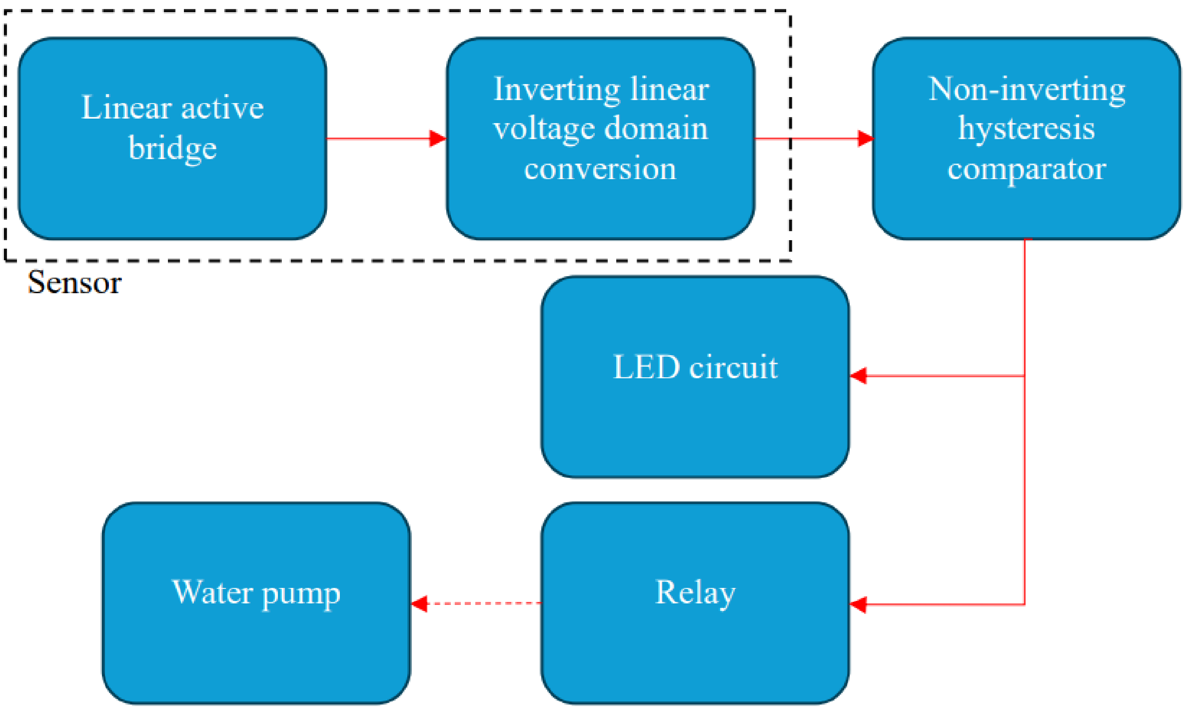

Figure 1 Block Diagram

3. System Components and Justification

Moisture Sensor

- **Component**: Variable Resistor (Rs)
- **Justification**: The soil moisture sensor is simulated using a variable resistor that changes resistance based on soil moisture levels.
- **Value**: 750kΩ (dry) to 100kΩ (wet)
- **Reason**: These values correspond to the resistance range of typical soil moisture sensors, allowing for accurate detection of moisture levels.

Resistors

Component: Fixed Resistors (R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, RLED)

- **Justification:** Resistors are used to create voltage dividers, set reference voltages, and limit current in the circuit.
- **Values and Reasons:**
  - **R1, R2, R3:** 1MΩ each - Form part of the voltage divider with the moisture sensor.
  - **R4:** 14kΩ - Sets a reference voltage in the circuit.
  - **R5:** 7.15kΩ - Part of the feedback network for the op-amp.
  - **R6:** 25.5kΩ - Sets a reference voltage in the circuit.
  - **R7:** 14.3kΩ - Sets a reference voltage in the circuit.
  - **R8:** 113kΩ - Sets a reference voltage in the circuit.
  - **R9:** 20kΩ - Part of the feedback network for the op-amp.
  - **R10:** 10.7kΩ - Lower threshold voltage divider.
  - **R11:** 13.7kΩ - Upper threshold voltage divider.
  - **R12:** 1.56kΩ - Base resistor for the transistor.
  - **RLED:** 280kΩ - Current limiting resistor for the LED.
  - **Relay:**400Ω- a resistor that will substitute the relay.

Operational Amplifier (Op-Amp)

- **Component**: AD822/AD
- **Justification**: Used for signal conditioning of the moisture sensor output.
- **Reason**: The AD822/AD is a low-power dual op-amp suitable for this application.

Comparator

- **Component**: AD822/AD
- **Justification**: Compares the conditioned sensor signal against set thresholds to control the irrigation system.
- **Reason**: The AD822/AD is a low-power dual comparator suitable for precise voltage comparison.

Transistor

- **Component**: 2N7000/FAI
- **Justification**: Acts as a switch to control the relay based on the comparator output.
- **Reason**: The 2N7000/FAI is a general-purpose NPN transistor with sufficient current handling for the relay coil.

Relay

- **Component**: Simulated by a resistor
- **Justification**: Controls the water pump based on the transistor's state.
- **Reason**: The relay can handle higher currents and voltages required to operate a water pump.

LED

- **Component**: Purple LED
- **Justification**: Provides a visual indication of the relay state.
- **Reason**: The LED color can be chosen based on preference, with a current limiting resistor to prevent damage.
- Because there isn`t a Purple LED in PSPICE, I modeled a diode in PSPICE MODEL EDITOR. The LEDs Datasheet can be found in the references.

Power Supply

- **Component**: 11V DC Power Supply
- **Justification**: Powers the entire circuit.
- **Reason**: Provides a stable voltage for consistent circuit operation.

4. Circuit Design and Calculations

In this section, I will explain the selection and calculation of each component in the circuit, including why choosing each value.

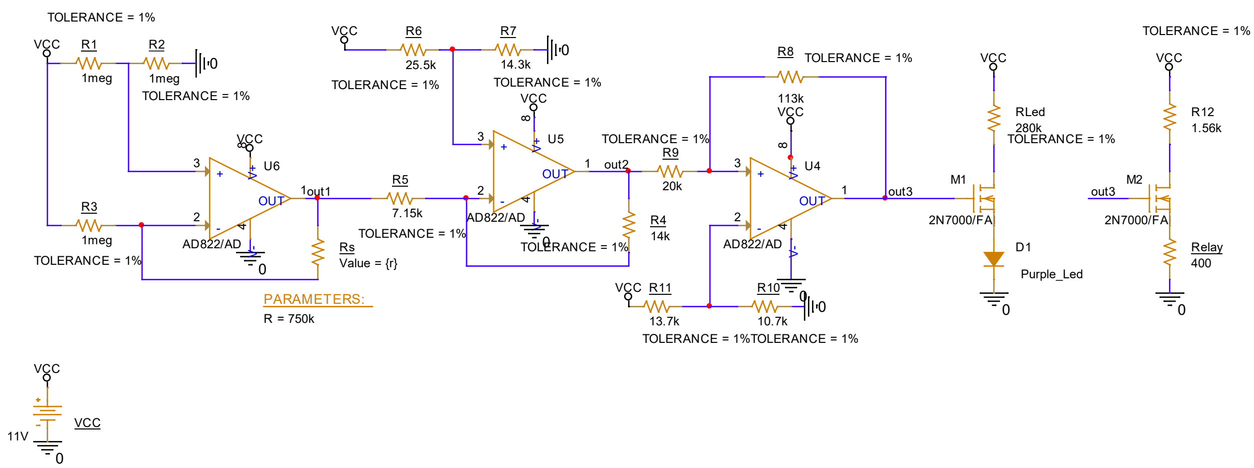

Figure 2: Complete Circuit

4.1Linear Active Bridge

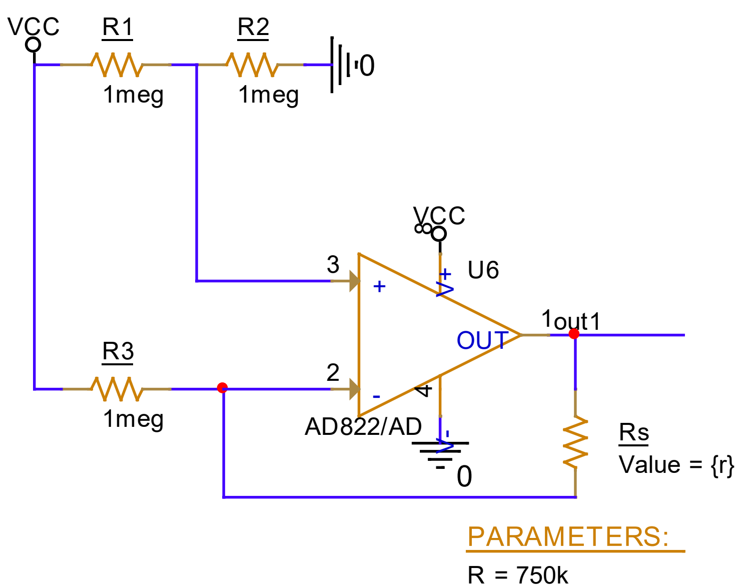

Figure 3: Linear Active Bridge

The active bridge is the component of the sensor that converts the variable resistance into a linear voltage variation. The suggested design for this block is illustrated in figure 3.

Firstly, by using the voltage divider equation, the expression of v+ can be deduced. Secondly, by using the superposition principle, the v- can be expressed.

$$v+=\frac{R1}{R1+R2}*Vcc$$

$$v-=\frac{Rs}{Rs+R3}*Vcc+\frac{R3}{R3+Rs}*Vout1$$

Based on the formulas mentioned above, we can calculate Vout1 min and Vout1 max.

R1, R2, R3 (1megΩ)

- **Reason for Selection**: This value is common in precision resistor networks in bridge configurations.
- **For Rs = 750kΩ(dry):**

$$\text{Vout1}_{\text{MIN}} = 5.5\text{V} - 5.5\text{V} \times \frac{750\text{K}\Omega}{1\text{M}\Omega} \approx 1.375\text{V}$$

- **For Rs=100kΩ (wet):**

$$\text{Vout1}_{\text{MAX}} = 5.5\text{V} - 5.5\text{V} \times \frac{100\text{K}\Omega}{1\text{M}\Omega} \approx 4.95\text{V}$$

4.2Inverting Linear Voltage Domain Converter (U1B - LM358)

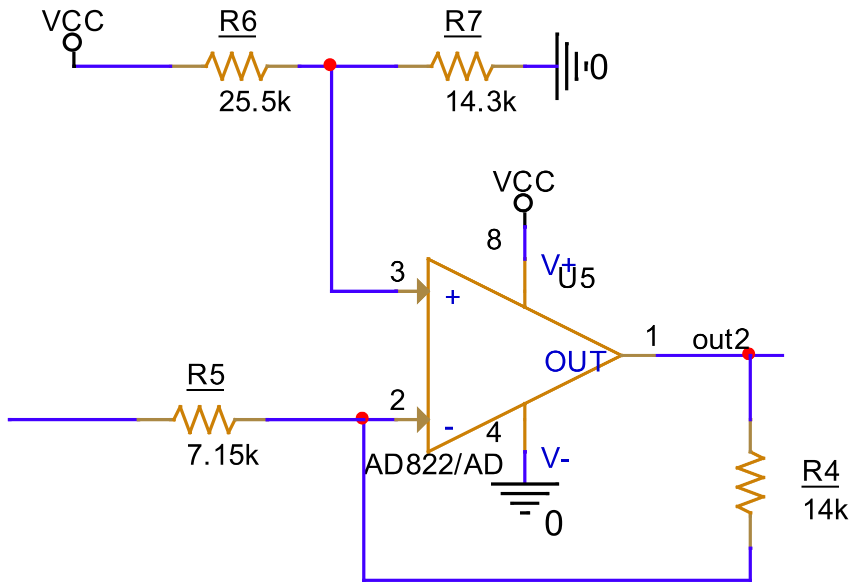

Figure 4: Inverting Linear Voltage Domain Converter

This block ensures that the sensor's output voltage ranges from 2V to 9V. The input voltage of this stage is Vout1 and a reference voltage is also used to achieve the appropriate output voltage swing. The circuit for this stage is shown in the figure.

$$v+= Vref = \frac{R6}{R6+R7} * Vcc$$

$$v-= \frac{R4}{R4+R5} * Vout1 + \frac{R5}{R4+R5} * Vout2$$

The ratio $\frac{R4}{R5}$ will be denoted with "m" for readability reasons.

$$Vout2 = (1+m) * Vref - m * Vout1 \Leftrightarrow \text{Vref} = \frac{Vout2 + m * Vout1}{1+m}$$

$$\begin{cases} Vref = \dfrac{2 + m * 4.95}{1+m} \\ Vref = \dfrac{9 + m * 1.375}{1+m} \end{cases} => \frac{2 + m * 4.95}{1+m} = \frac{9 + m * 1.375}{1+m} <=>$$

$$m * (4{,}95 - 1{,}375) = 7 <=> m = 1.958$$

$$=> Vref = 3.952V$$

$$\frac{R6}{R6 + R7} * Vcc = 3.952V <=> \frac{R6}{R6 + R7} = \frac{3.952V}{11V}$$

The values for the R6 and R7 resistances were then found using an online standard resistor value calculator.

$$R6 = 25.5k\Omega \quad R7 = 14.3k\Omega$$

Finally, m was replaced with $\frac{R4}{R5}$ and then the same online calculator was used to find standard E96 values.

$$R4 = 14k\Omega \quad R5 = 7.15k\Omega$$

4.3Non-inverting hysteresis comparator

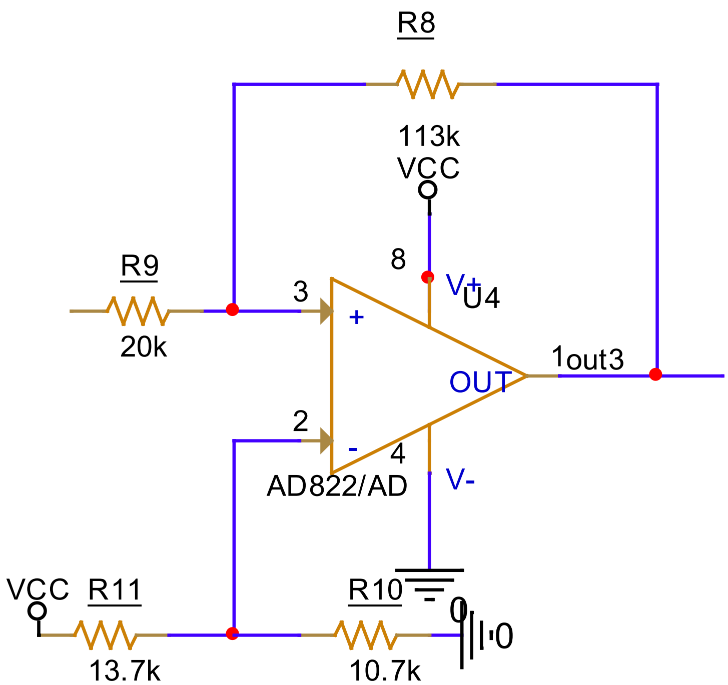

Figure 5: Non-inverting hysteresis comparator

In the following section, we will discuss the switching stage of the circuit. This block is responsible for delivering a “HIGH” voltage level (11V) when the low humidity threshold is encountered and a “LOW” voltage level (0V) when the humidity reaches the high threshold.

This effect was achieved using an operational amplifier with positive feedback. The input of this stage is denoted as Vout2, and the output is Vout3. Additionally, a reference voltage, denoted as Voff, was applied at the inverting input V-. Without this reference voltage, the threshold voltages would have been symmetrical around the mean value of the supply voltages 11V and 0V powering the operational amplifier.

The required threshold values were determined based on the relationship between the humidity levels and the varying resistance of the sensor.

In order to size the threshold voltages, it must be taken into account that Vout3 has only two stable states.

$$\mathrm{Vout3} = \{0\mathrm{v}, 11\mathrm{V}\}$$

By using the formulas for the non-inverting input voltage v+ and the inverting input voltage v- , an expression of the differential voltage can be found.

$$v+= \frac{R8}{R8+R9} * Vout2 + \frac{R9}{R8+R9} * Vout3$$

$$v-= Voff = \frac{R10}{R10+R11} * Vcc$$

$$vD = v + \ -v-= \frac{R8}{R8+R9} * Vout2 + \frac{R9}{R8+R9} * Vout3 - Voff$$

$$5.67V - 1.95V = \left(1+\frac{R9}{R8}\right) * Voff - \left(1+\frac{R9}{R8}\right) * Voff + \frac{R9}{R8} * 11V <=>$$

$$\frac{R9}{R8} = \frac{1.95}{11} = 0.117 => R8 = 113k\Omega \ \ R9 = 20k\Omega$$

$$5.67V = (1+0.117) * \frac{R10}{R10+R11} * 11V => R10 = 10.7k\Omega \ \ R11 = 13.7k\Omega$$

4.4Led Circuit

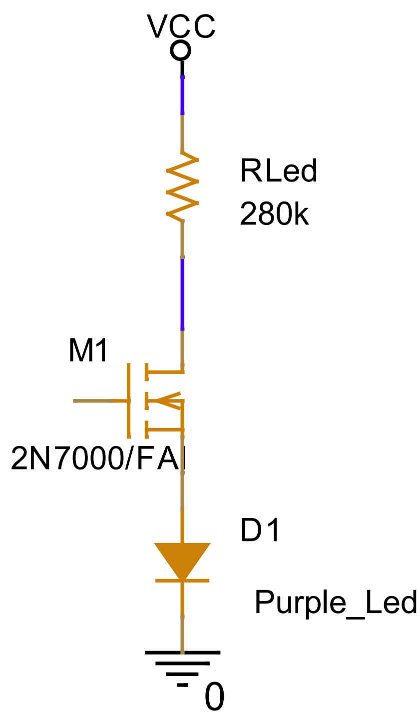

Figure 6: Led Circuit

The purpose of the LED is to indicate when the water pump is activated, meaning the LED lights up when the output voltage of the comparator is 11V.

The resistance for the LED was calculated using the forward voltage(4V) and forward current(25mA) provided in the LED Data Sheet.

$$11V - 4V = 7V$$

$$RLed \frac{7V}{25mA} = 280\Omega$$

4.5Relay Circuit

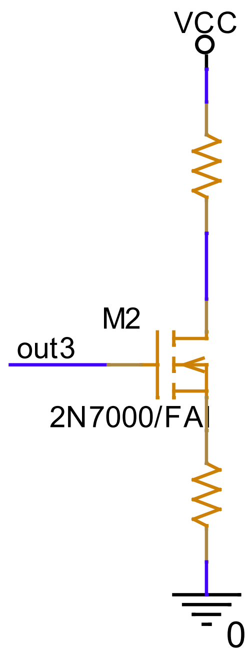

Figure 7 Relay Circuit

Typical 12V relays have coil resistances ranging from around 200Ω to 600Ω. For this example, let's choose a relay with a coil resistance of 400Ω.

$$Icoil = \frac{Vcoil}{Rcoil} = \frac{11V}{400\Omega} = 27.5mA$$

$$\text{IB} = \frac{Icoil}{hFE} = \frac{27.5mA}{100} = 0.275mA$$

$$RB = \frac{Vout3 - VBE}{IB} = \frac{5V - 0.7V}{2.75mA} = 1.56k\Omega$$

$$R12 = 1.56k\Omega$$

5. List of Components and Prices(Bill of Materials)

Component	Part Number	Quantity	Unit Price (€)	Total Price (€)	Supplier
Resistor	1MΩ	3	0.1	0.3	Digi-key
Resistor	7.15kΩ	1	0.1	0.1	Digi-key
Resistor	14kΩ	2	0.1	0.2	Digi-key
Resistor	10.7kΩ	1	0.1	0.1	Digi-key
Resistor	13.7kΩ	1	0.1	0.1	Digi-key
Resistor	20kΩ	1	0.1	0.1	Digi-key
Resistor	25.5kΩ	1	0.1	0.1	Digi-key
Resistor	56kΩ	1	0.1	0.1	Digi-key
Resistor	75kΩ	1	0.1	0.1	Digi-key
Resistor	113kΩ	1	0.1	0.1	Digi-key
Resistor	14.3kΩ	1	0.1	0.1	Digi-key
Resistor	1.56kΩ	1	0.1	0.1	Digi-key
Resistor	280Ω	1	0.1	0.1	Digi-key
Relay Resistor	400Ω	1	0.1	0.1	Digi-key
Op-Amp	AD822	2	0.5	1	Digi-key
Comparator	AD822	1	0.5	0.5	Digi-key
Transistor	2N7000/FAI	2	0.2	0.4	Digi-key
LED	Purple LED	1	0.2	0.2	Digi-key
Power Supply	11V DC	1	10.0	10	Amazon
Total Price				13.8	

6. Simulation and Analysis

6.1 Simulation

This section is meant to serve as an example of how the circuit works as a block assembly. Furthermore, a few of the simulations that have been run attempt to confirm that the circuit is accurate when component tolerance is taken into account.

6.1.1 DC SWEEP Analysis

The DC sweep analysis of a global parameter is the most significant simulation for this circuit. To conduct this analysis, the operating point of the circuit is determined for a range of parameter values within a predetermined interval.

First, using the simulation set shown in figure 8, the resistance of Rs will be varied linearly between 100kΩ and 750kΩ. This step is necessary to determine whether the sensor's output or Vout2, varies linearly between 2 and 9 volts. The graphic that results is shown in figure 9.

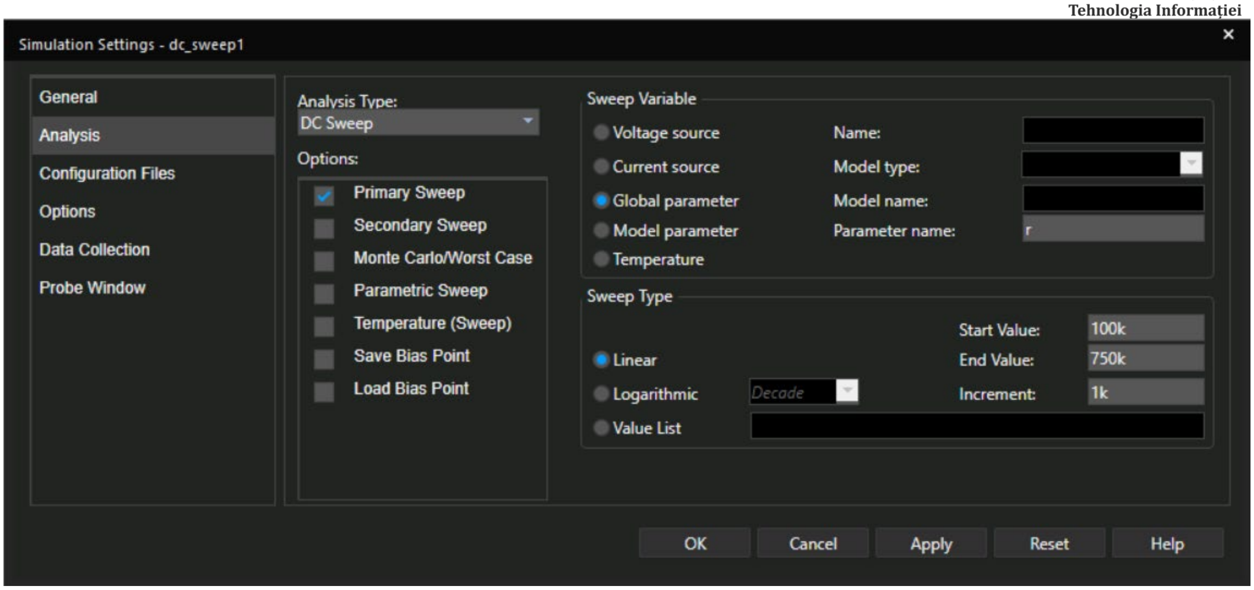

Figure 8 DC Sweep Analysis Settings

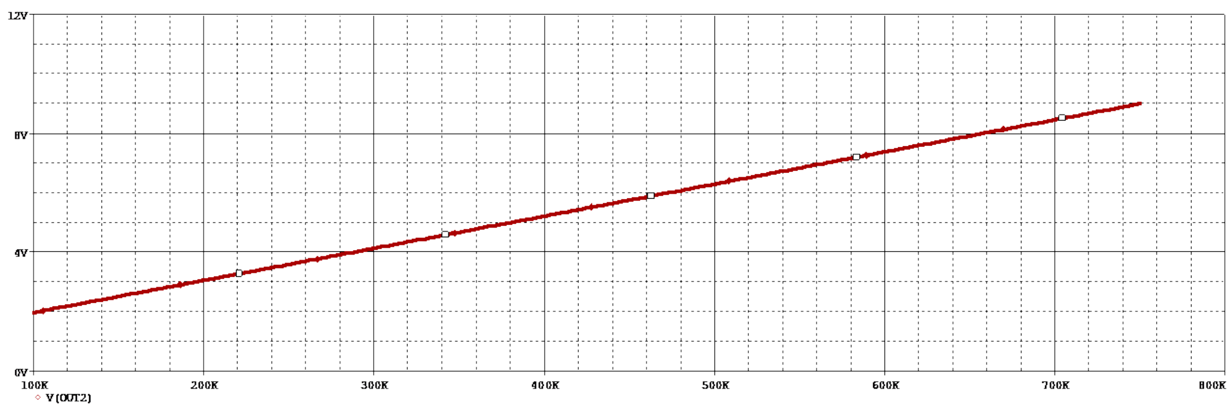

Figure 9 Vout2 as a function of Rs plot

The minimum Max(V(out2) and the highest Min(V(out2) data were examined since the precise bounds of Vout2 cannot be distinguished on the previous display. The findings are shown in figure 10.

Trace Name	Y1	Y2
X Values	749.000K	100.000K
V(OUT2)	8.9791	1.9613

Figure 10

$$humidity[\%] = \frac{(750k - Rs) * (90 - 20)}{750k - 100k} + 20$$

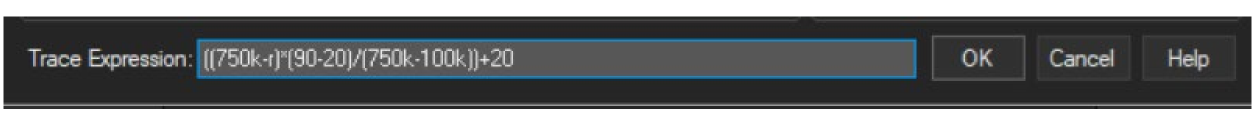

Figure 11 Expression to display Humidity[%] on the X-axis

To show the fluctuation of Vout2 with respect to humidity, the formula above was used in place of the X-axis variable in the previous plot in figure 8. The outcome is shown in figure 12

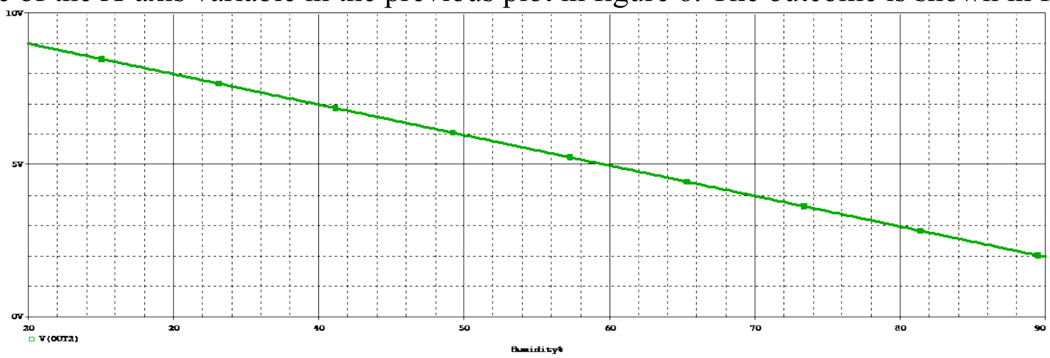

Figure 12

The next step was to confirm that the comparator stage's output voltage, which is 11V, is "HIGH" when the humidity drops to 55%. The identical simulation as shown in figure 8 was applied in this context. The outcome is shown in figure 13.

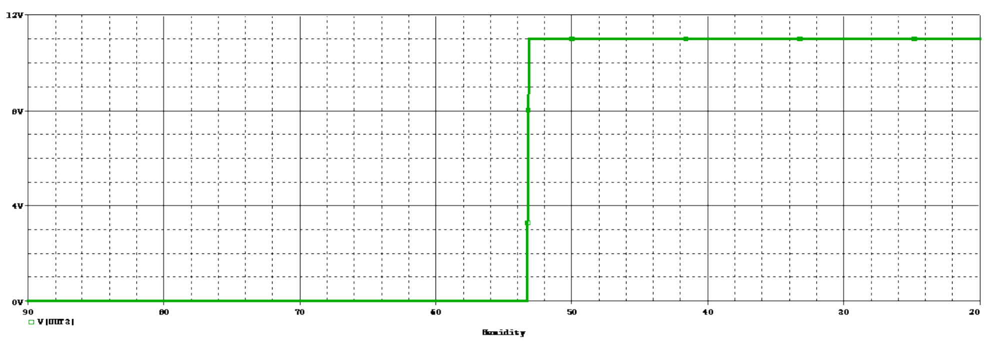

Figure 13

In order to see if the comparator stage's output changes from "HIGH" to "LOW" (11V to 0V) at 75% soil moisture, the simulation profile was adjusted to increase humidity from 55% to 85%. The next two figures show the revised simulation profile and the plot that was produced.

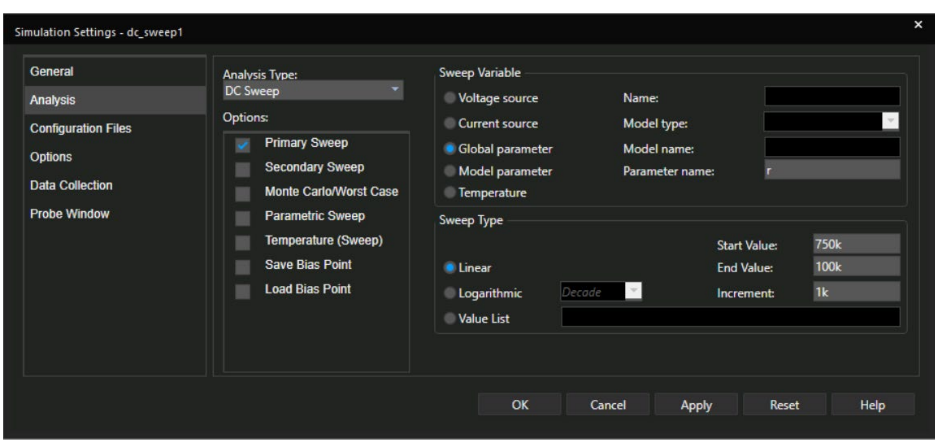

Figure 14 DC Sweep Analysis Settings

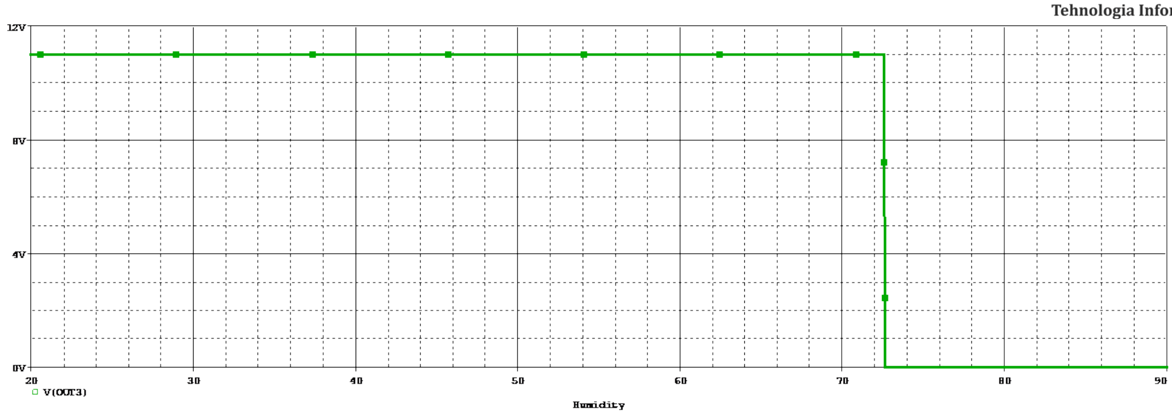

Figure 15

6.1.2 Worst Case

The worst-case analysis was carried out to ascertain the extent to which the output voltage range of the sensor and the threshold levels of the hysteresis comparator can be affected by the divergence of the resistances within the tolerance interval from their nominal values. The simulation profile is defined in figure 16, and Vout2 and Vout3 are presented as functions of humidity in figures 17 and 18. The nominal value situation is shown by the green plots, and the worst case is shown by the red ones.

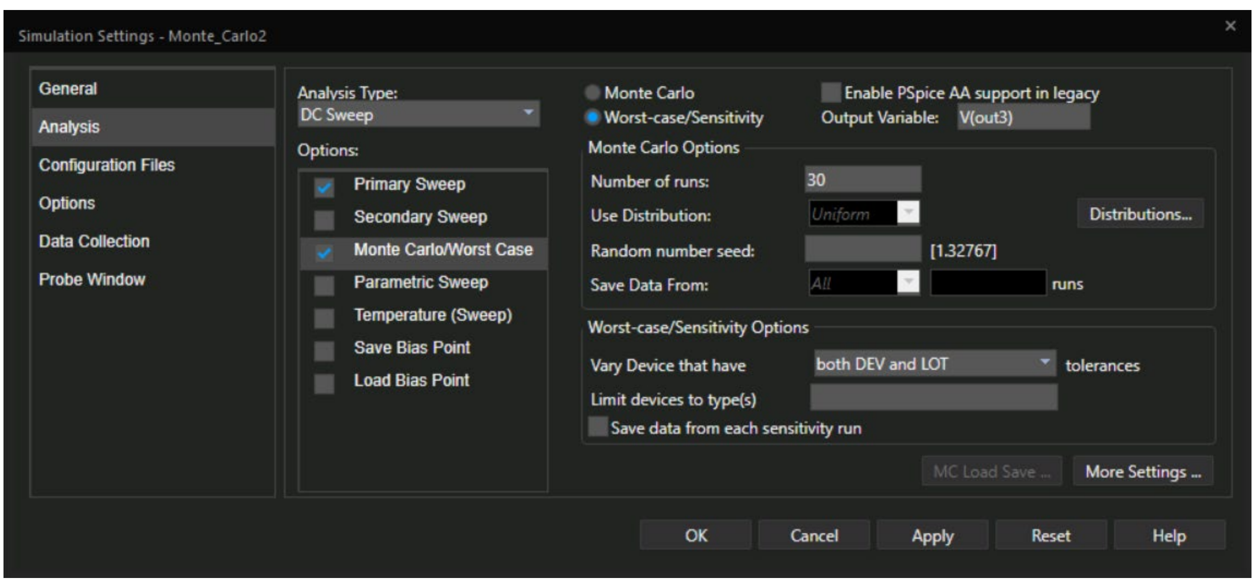

Figure 16 Worst Case Analysis Settings

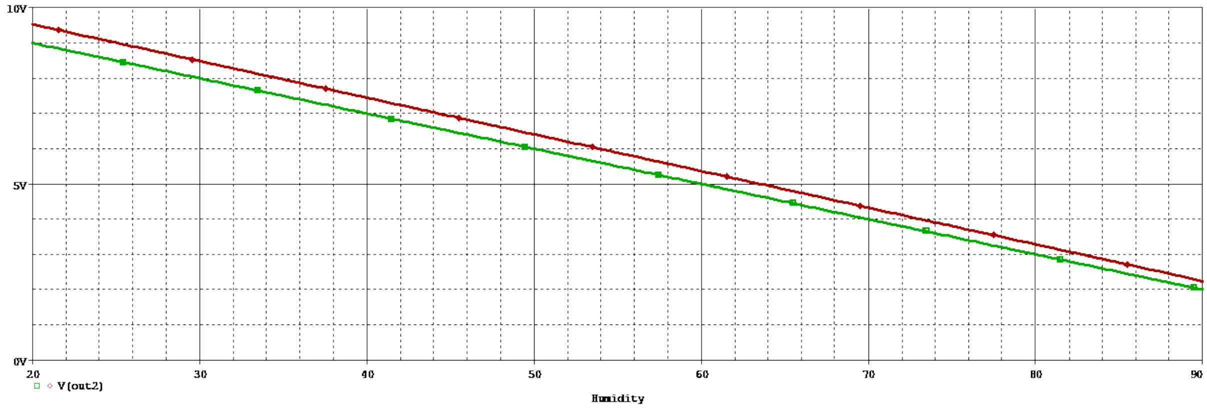

Figure 17

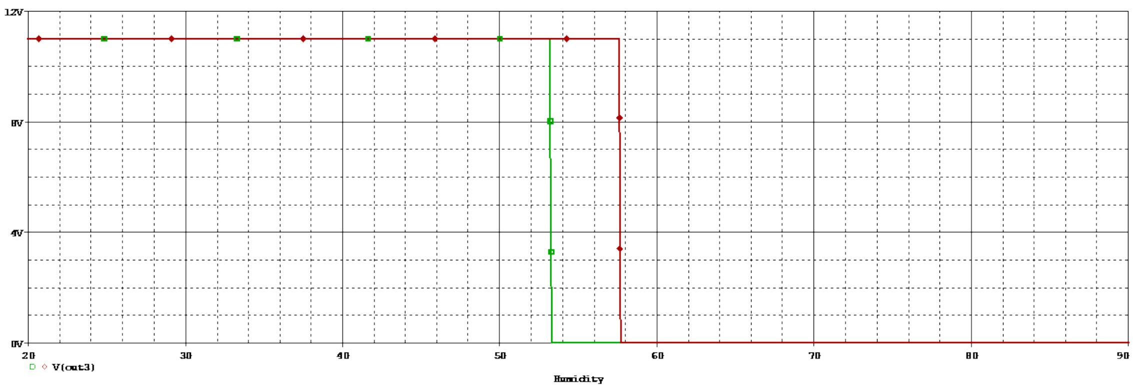

Figure 18

By measuring the plots, it was determined that the Vout2 voltage swing is [2.24V, 9.506V] in the worst case.

Trace Color	Trace Name	Y1	Y2	Y1 - Y2
	X Values	748.000K	100.000K	648.000K
CURSOR 1,2	V(out2)	8.9765	1.9983	6.9781
	V(out2)	9.506	2.2432	7.2629

Figure 19

6.1.3 Monte Carlo

The statistical histogram showing the effects of component tolerance on the sensor's output voltage swing was produced by the application of Monte-Carlo analysis. Figure 20, which shows the simulation profile, is followed by figure 21 & 22 which shows the variation of the sustained humidity range on one panel and the histogram of the highest level of Vout2.

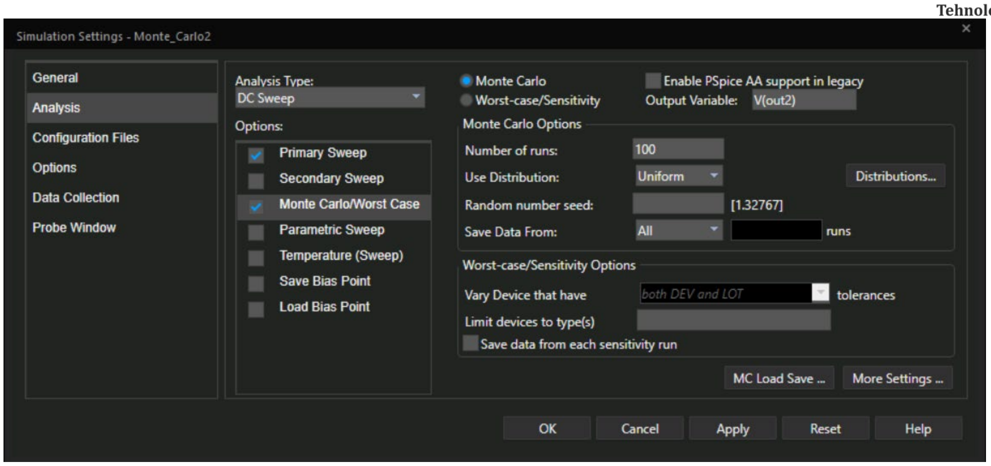

Figure 20 Monte Carlo Analysis Settings

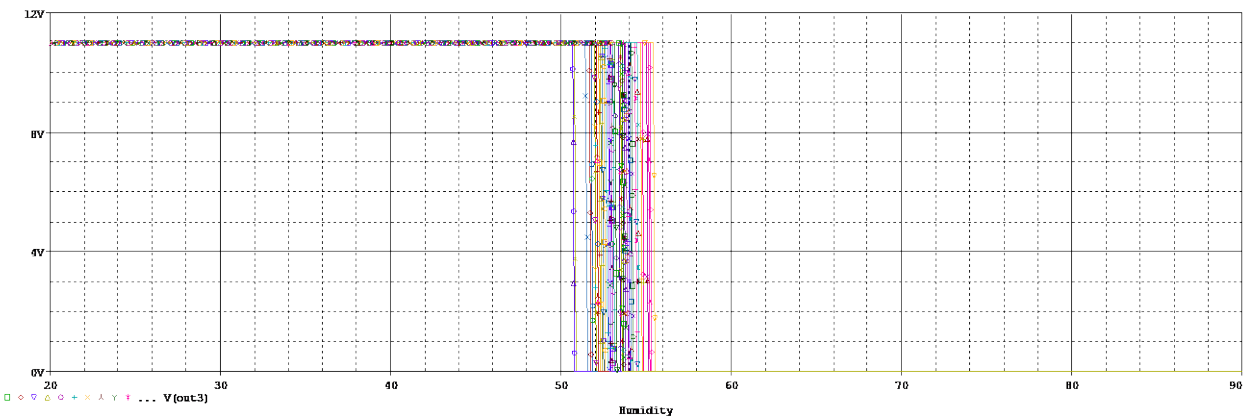

Figure 21

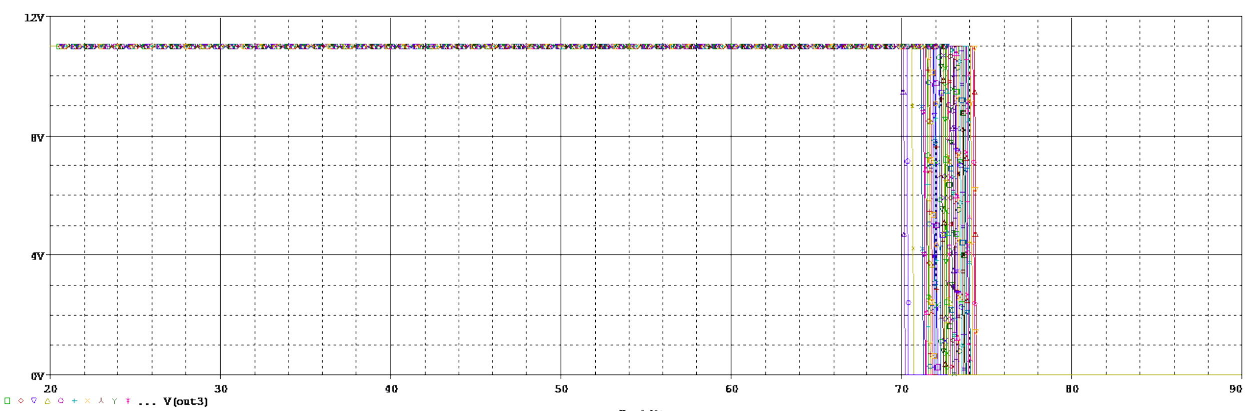

Figure 22

6.1.4 Temperature

It is unnecessary to do a temperature analysis because the circuit is intended for indoor use, where the typical temperature range is between 10°C and 35°C. Consequently, there is little difference in the measured values' temperature.

7. Conclusion

This project effectively showcases the design and execution of an automated irrigation system utilizing a soil moisture sensor, operational amplifiers, and a hysteresis-based comparator. The technology is engineered to sustain ideal soil moisture levels, guaranteeing that plants receive sufficient water without the need for human intervention. Utilizing a soil moisture sensor enables the system to precisely measure moisture levels, while the op-amp arrangement amplifies and prepares the sensor output for comparison. The inclusion of hysteresis in the comparator enhances stability by preventing quick switching of the relay, hence ensuring dependable functioning. The transistor and relay control system efficiently regulates the irrigation pump, activating or deactivating it according to the soil moisture conditions.

By doing meticulous design calculations, carefully selecting components, and conducting rigorous simulations, the project confirms the practicality of the automated irrigation system. Utilizing PSpice for simulation, which encompasses DC sweep and Monte Carlo analysis, guarantees that the system functions as anticipated under different conditions and component tolerances. In summary, the technology exhibits substantial promise in terms of water conservation and labor reduction in agricultural and gardening contexts.

8.Future Work

Although the current project effectively achieves its goals, there are other aspects that can be enhanced and expanded upon in the future:

1. IoT Integration: Improve the system by using the Internet of Things (IoT) to facilitate remote monitoring and control.
- Specifics: Integrate wireless communication modules, such as Wi-Fi and Bluetooth, to establish connectivity between the system and a central hub or cloud platform. Create a mobile application or web interface to enable live monitoring and management of the irrigation system.

2. Enhanced Sensor Integration: Enhance the precision and dependability of the system by including supplementary sensors.
-Install temperature, humidity, and light intensity sensors to obtain a thorough assessment of environmental conditions. Apply data fusion methodologies to enhance irrigation scheduling by considering numerous aspects.

3. Power Efficiency: Enhance the power efficiency of the system to cater to remote or off-grid areas.
-Install photovoltaic panels and lithium-ion batteries to provide energy for the system. Enhance the circuit design to minimize power usage by employing energy-efficient components and incorporating sleep modes.

4. Scalability: Enhance the system's capacity to handle greater geographical areas or numerous zones.
Create a modular system architecture that facilitates seamless scalability. Install zone-specific controls and valves to autonomously regulate irrigation for distinct locations.

5. Data Analytics and Machine Learning: Employ sophisticated data analytics and machine learning techniques to forecast irrigation requirements and enhance water utilization.
-Gather historical data regarding the levels of moisture in the soil, meteorological conditions, and the growth of plants. Utilize machine learning algorithms to forecast future irrigation needs and adapt the system settings accordingly.

6. User Interface and Alerts: Improve the user experience by offering user-friendly interfaces and informative notifications.
- Create an intuitive dashboard that presents up-to-date information and the current state of the system in a way that is easy for users to understand and interact with. Integrate alert methods, such as SMS or email, to promptly inform consumers about crucial situations or system failures.

By exploring these prospective fields, the automated irrigation system can be enhanced to become a more advanced, dependable, and user-friendly solution, while promoting sustainable agriculture and effective water management techniques.

9. References

1. **Datasheets:**
   - LM358 Dual Operational Amplifier Datasheet
     - Texas Instruments. "LM358 Dual Operational Amplifier."
     - Documentation Link
   - LM393 Dual Comparator Datasheet
     - Texas Instruments. "LM393 Low-Power Dual Comparator."
     - Documentation Link
   - 2N2222A NPN Transistor Datasheet
     - ON Semiconductor. "2N2222A NPN Switching Transistor."
     - Documentation Link
   - UV/UVA 400nm Purple LED 5mm Clear Lens
     - Documentation Link

2. **Component Suppliers:**
    - Digi-Key Electronics. Digi-Key
    - Amazon for the 11V DC Power Supply. Amazon
3. **Design Principles:**
    - Sedra, Adel S., and Kenneth C. Smith. "Microelectronic Circuits." Oxford University Press, 2014.
        - This textbook provides fundamental principles on operational amplifiers, comparators, and transistor-based circuits.
    - Horowitz, Paul, and Winfield Hill. "The Art of Electronics." Cambridge University Press, 2015.
        - A comprehensive reference for electronic circuit design
4. **Simulation Techniques:**
    - OrCAD PSpice Simulation. Cadence OrCAD
        - Official documentation and tutorials on performing DC sweep, transient analysis, and Monte Carlo analysis in PSpice.
5. **General Electronics References:**
    - "Electronic Devices and Circuit Theory" by Robert L. Boylestad and Louis Nashelsky. Pearson, 2012.
        - This book covers the theory and application of electronic devices and circuits, including detailed information on components used in the project.
6. **Online Resources:**
    - Texas Instruments E2E Community. TI E2E
        - An online community forum where engineers discuss and solve problems related to Texas Instruments components.
    - Electronics Tutorials. Electronics Tutorials
        - A comprehensive website that provides tutorials on various electronic components and circuit design principles.
    - Computer Aided Design Soil moisture control circuit-Medeleanu Ștefan
        - One of my colleagues' documentations that really helped me understand and complete this project.